

Boris Bonev

SENIOR RESEARCH SCIENTIST · NVIDIA

✉ bonevbs@gmail.com | 🌐 bonevbs.github.io | 🎓 Google Scholar

Senior Research Scientist specializing in AI for Science (AI4Science), bridging **rigorous numerical methods** and **scalable machine learning**. Pioneer of Spherical Fourier Neural Operators and the FourCastNet weather model family. Proven track record of developing principled AI methods at **operational scale** (NOAA, ECMWF) on **1,000+ GPU** clusters.

Research and Work Experience

NVIDIA

Zurich, Switzerland

SENIOR RESEARCH SCIENTIST

2023 - Present

- Led the development of **frontier AI weather models** (FourCastNet 2/3) enabling unprecedented **10'000 member ensembles**
- Formulated key supporting innovations such as **Spherical Fourier Neural Operators** and **local integral neural operators**
- Research operationalized by major agencies (NOAA, AWS, ECMWF, MetDesk), transforming global weather forecasting
- Drove the development of **standard tools** and **frameworks** adopted by researchers and engineers world-wide
- Initiated and directed collaborative research agendas with UC Berkeley, Lawrence Berkeley Labs, Caltech and Allen Institute
- Mentored and supervised 5 interns and colleagues guiding their research to publication at premier venues
- Research featured in White House PCAST presentation (Anima Anandkumar, 2023)

Swiss Federal Institute of Technology Lausanne (EPFL)

Lausanne, Switzerland

PHD IN APPLIED MATHEMATICS

2017-2021

- Developed a scalable, sparse direct solver using hierarchical matrices to achieve quasi-linear complexity
- Established a state-of-the-art discontinuous Galerkin method for the spherical shallow water equations
- Served as principal teaching assistant during seven semesters for analysis, probability and linear algebra courses

TU Munich

Munich, Germany

RESEARCH ASSISTANT IN MACHINE LEARNING

2016 - 2017

- Research and design of machine learning algorithms for fluid simulations
- Served as principal teaching assistant for an introductory course in machine learning

Education

Swiss Federal Institute of Technology Lausanne (EPFL)

Lausanne, Switzerland

PHD IN APPLIED MATHEMATICS

2017-2021

- Focus on randomized linear algebra and numerical methods for hyperbolic PDEs
- Thesis: **Efficient algorithms for wave problems**
- **Advisor:** Jan S. Hesthaven, **Committee:** Daniel Kressner, Per-Gunnar Martinsson, and Remi Abgrall

University of Stuttgart

Stuttgart, Germany

B.S. IN PHYSICS

2011-2015

- Completed in parallel with Aerospace Engineering degree
- Focus: Theoretical Physics, Statistical Physics and Computational Physics

University of Stuttgart

Stuttgart, Germany

M.S. AND B.S. IN AEROSPACE ENGINEERING

2009-2015

- Specialization: Numerical and Mathematical Modelling

Publications

Most recent publications via [Google scholar](#). Equal contribution indicated by *.

CONFERENCE AND JOURNAL PUBLICATIONS

1. **Boris Bonev***, Max Rietmann*, Andrea Paris, Andrea Carpentieri, and Thorsten Kurth*. "Attention on the Sphere." In Neural Information Processing Systems (NeurIPS), 2025.
2. Ankur Mahesh, William D. Collins, **Boris Bonev**, Noah Brenowitz, Yair Cohen, Joshua Elms, Peter Harrington, et al. "Huge ensembles-Part 1: Design of ensemble weather forecasts using spherical Fourier neural operators." Geoscientific Model Development 18, no. 17 (2025): 5575-5603.

3. Ankur Mahesh, William D. Collins, **Boris Bonev**, Noah Brenowitz, Yair Cohen, Peter Harrington, Karthik Kashinath, et al. "Huge ensembles–Part 2: Properties of a huge ensemble of hindcasts generated with spherical Fourier neural operators." *Geoscientific Model Development* 18, no. 17 (2025): 5605-5633.
4. Noah D. Brenowitz, Yair Cohen, Jaideep Pathak, Ankur Mahesh, **Boris Bonev**, Thorsten Kurth, Dale R. Durran, Peter Harrington, and Michael S. Pritchard. "A practical probabilistic benchmark for ai weather models." *Geophysical Research Letters* 52, no. 7 (2025): e2024GL113656.
5. Miguel Liu-Schiaffini*, Julius Berner*, **Boris Bonev***, Thorsten Kurth, Kamyar Azizzadenesheli, and Anima Anandkumar. "Neural Operators with Localized Integral and Differential Kernels." In *International Conference on Machine Learning*, pp. 32576-32594. PMLR, 2024.
6. James PC Duncan, Emily Wu, Jean-Christophe Golaz, Peter M. Caldwell, Oliver Watt-Meyer, Spencer K. Clark, **Boris Bonev**, et al. "Application of the AI2 Climate Emulator to E3SMv2's global atmosphere model, with a focus on precipitation fidelity." *Journal of Geophysical Research: Machine Learning and Computation* 1, no. 3 (2024).
7. Renbo Tu, Colin White, Jean Kossaifi, **Boris Bonev**, Gennady Pekhimenko, Kamyar Azizzadenesheli, and Anima Anandkumar. "Guaranteed Approximation Bounds for Mixed-Precision Neural Operators." In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024.
8. Md Ashiqur Rahman, Robert Joseph George, Manal Elleithy, Daniel V. Lebovici, Zongyi Li, **Boris Bonev**, Colin White, Julius Berner, Raymond A. Yeh, Jean Kossaifi, and Kamyar Azizzadenesheli. "Pretraining codomain attention neural operators for solving multiphysics pdes." *Advances in Neural Information Processing Systems* 37 (2024): 104035-64.
9. Matthias Karlbauer, Noah Cresswell-Clay, Dale R. Durran, Raul A. Moreno, Thorsten Kurth, **Boris Bonev**, Noah Brenowitz, and Martin V. Butz. "Advancing parsimonious deep learning weather prediction using the HEALPix mesh." *Journal of Advances in Modeling Earth Systems* 16, no. 8 (2024): e2023MS004021.
10. **Boris Bonev**, Thorsten Kurth, Christian Hundt, Jaideep Pathak, Maximilian Baust, Karthik Kashinath, and Anima Anandkumar. "Spherical Fourier Neural Operators: Learning Stable Dynamics on the Sphere." In *International Conference on Machine Learning*, pp. 2806-2823. PMLR, 2023.
Oral (top 8.5% of accepted papers).
11. **Boris Bonev**, and Jan S. Hesthaven. "A hierarchical preconditioner for wave problems in quasilinear complexity." *SIAM Journal on Scientific Computing* 44, no. 1 (2022): A198-229.
12. Mahya Hajihassanpour, **Boris Bonev**, and Jan S. Hesthaven. "A comparative study of earthquake source models in high-order accurate tsunami simulations." *Ocean Modelling* (2019).
13. **Boris Bonev**, Jan S. Hesthaven, Francis X. Giraldo, and Michal A. Kopera. "Discontinuous Galerkin scheme for the spherical shallow water equations with applications to tsunami modeling and prediction." *Journal of Computational Physics* 362 (2018): 425-448.
14. Lukas Prantl, **Boris Bonev**, and Nils Thuerey. "Generating liquid simulations with deformation-aware neural networks." *International Conference on Learning Representations* (2017).

PREPRINTS AND UNDER SUBMISSION

15. Jean Kossaifi, Nikola Kovachki, Morteza Mardani, Daniel Lebovici, Suman Ravuri, Ira Shokar, Edoardo Calvello, Mohammad Shoaib Abbas, Peter Harrington, Ashay Subramaniam, Noah Brenowitz, **Boris Bonev**, Wonmin Byeon, Karsten Kreis, Dale Durran, Arash Vahdat, Mike Pritchard and Jan Kautz. "Demystifying Data-Driven Probabilistic Medium-Range Weather Forecasting." preprint (2026).
16. Kwesi T. Quagraine, William D. Collins, Ankur Mahesh, **Boris Bonev**, Karthik Kashinath, Thorsten Kurth, Michael S. Pritchard, Shashank Subramanian. "Huge Ensembles Heatwave Forecasting over North Africa: Case Study of 3 Low-Likelihood, High-Impact Events." Under review at *Journal of Advances in Modeling Earth Systems* (2025).
17. **Boris Bonev***, Thorsten Kurth*, Ankur Mahesh, Mauro Bisson, Jean Kossaifi, Karthik Kashinath, Anima Anandkumar, William D. Collins, Michael S. Pritchard, and Alexander Keller. "FourCastNet 3: A geometric approach to probabilistic machine-learning weather forecasting at scale." *arXiv preprint arXiv:2507.12144* (2025).
18. Jipeng Wang, Youssef Aborahama, Abdullah Khokhar, Yinxu Zhang, Chen Wang, Kaustav Sastry, Julius Berner, Yuankai Luo, **Boris Bonev**, Zongyi Li, and Kamyar Azizzadenesheli. "Accelerating 3D Photoacoustic Computed Tomography with End-to-End Physics-Aware Neural Operators." *arXiv preprint arXiv:2509.09894* (2025).
19. Zhaoyang Dai, Mingxi Yin, Xinyi Chen, Andrea Carpentieri, Jussi Leinonen, **Boris Bonev**, Chengyuan Zhong, Thorsten Kurth, Jie Sun, Rama Cherukuri, and Yingjie Zhang. "Data-driven solar forecasting enables near-optimal economic decisions." *arXiv preprint arXiv:2509.06925* (2025).
20. Julius Berner, Miguel Liu-Schiaffini, Jean Kossaifi, Valentin Duruisseaux, **Boris Bonev**, Kamyar Azizzadenesheli, Anima Anandkumar. "Principled Approaches for Extending Neural Architectures to Function Spaces for Operator Learning." *arXiv*

preprint arXiv:2506.10973 (2025).

21. Jean Kossaifi, Nikola Kovachki, Zongyi Li, David Pitt, Miguel Liu-Schiaffini, Valentin Duruisseaux, **Boris Bonev**, et al. "A library for learning neural operators." arXiv preprint arXiv:2412.10354 (2024).
22. Andrea Carpentieri, Jussi Leinonen, James Adie, **Boris Bonev**, Doris Folini, and Fabia Hariri. "Data-driven Surface Solar Irradiance Estimation using Neural Operators at Global Scale." arXiv preprint arXiv:2411.08843 (2024).
23. Jussi Leinonen, **Boris Bonev**, Thorsten Kurth, and Yair Cohen. "Modulated adaptive fourier neural operators for temporal interpolation of weather forecasts." arXiv preprint arXiv:2410.18904 (2024).
24. Shoaib Ahmed Siddiqui, Jean Kossaifi, **Boris Bonev**, Christopher Choy, Jianren Kautz, David Krueger, and Kamyar Azizzadenesheli. "Exploring the design space of deep-learning-based weather forecasting systems." arXiv preprint arXiv:2410.07472 (2024).
25. Chen Wang, Michael S. Pritchard, Noah Brenowitz, Yair Cohen, **Boris Bonev**, Thorsten Kurth, Dale Durran, and Jaideep Pathak. "Coupled ocean-atmosphere dynamics in a machine learning earth system model." arXiv preprint arXiv:2406.08632 (2024).
26. Oliver Watt-Meyer, Gideon Dresdner, Jeremy McGibbon, Spencer K. Clark, Brian Henn, James Duncan, **Boris Bonev**, et al. "ACE: A fast, skillful learned global atmospheric model for climate prediction." arXiv preprint arXiv:2310.02074 (2023).

IN PREPARATION

27. Thorsten Kurth, Max Rietmann, Mauro Bisson, Alberto Carpentieri, Andrea Paris, Jean Kossaifi, Christian Hundt, and **Boris Bonev**. "A library for differentiable signal processing and machine learning on the sphere." In preparation (2026).
28. Francesco Leonardi, **Boris Bonev**, and Kaspar Riesen. "MARA: Continuous SE(3)-Equivariant Attention for Molecular Force Fields." In preparation (2026).
29. Alistair White, Valentin Duruisseaux, **Boris Bonev**, Kamyar Azizzadenesheli, Anima Anandkumar, Niklas Boers. "Enforcing Conservation Laws in Neural Operators for Earth System Modeling." In preparation (2026).
30. Abdoul R. Zeba, William D. Collins, Ankur Mahesh, **Boris Bonev**, Karthik Kashinath, Thorsten Kurth, Michael Pritchard, and Shashank Subramanian "Anomalous Diffusion of Tropical Cyclones Observed in Huge Ensembles of Hindcasts." In preparation (2026).
31. Mark Risser, Ankur Mahesh, Joshua North, William D. Collins, **Boris Bonev**, Karthik Kashinath, Thorsten Kurt, Michael S. Pritchard, Shashank Subramanian "Surface Temperature Extremes Produced by Huge Ensembles of Hindcasts of Summer 2023." In preparation (2026).

Presentations

06/2026	Invited talk. <i>Operator Learning for Hyperbolic PDEs and its Applications in Weather and Climate Prediction.</i> SciCADE 2026	Edinburgh, United Kingdom
05/2026	Invited talk. <i>FourCastNet 3: A geometric approach to probabilistic machine-learning weather forecasting at scale.</i> EGU General Assembly	Vienna, Austria
03/2026	Invited talk. <i>Harnessing Huge Ensembles for Data-Driven Extreme Weather Prediction.</i> SIAM Uncertainty Quantification (UQ)	Minnesota, United States
03/2026	Invited talk. <i>Operator Learning for Hyperbolic PDEs and its Applications in Weather and Climate Prediction.</i> 96th Annual Meeting of the International Association for Applied Mathematics and Mechanics (GAMM)	Stuttgart, Germany
07/2025	Invited talk. <i>FourCastNet 3: A geometric approach to probabilistic machine-learning weather forecasting at scale.</i> ECMWF	virtual
06/2025	Invited talk. <i>FourCastNet 3: Advancing probabilistic weather forecast through machine learning at scale .</i> Platform for Advanced Scientific Computing (PASC) 2025	Brugg, Switzerland
06/2024	Contributed talk. <i>Advancing Probabilistic Weather Forecast through Machine Learning at Scale.</i> EXCLAIM workshop, ETH Zurich	Zurich, Switzerland

06/2025	Invited talk. <i>A principled approach to probabilistic machine-learning weather forecast at scale.</i> INRIA/CEA/EDF summerschool	Palaiseau, France
11/2024	Invited talk. <i>Digital Twins of our Planet in the Era of Machine Learning.</i> Bridging Modelling and Data Sciences, Swiss Geoscience Symposium	Lausanne, Switzerland
07/2024	Contributed talk. <i>Local Integral and Differential Neural Operators.</i> Spotlight at AI for Science workshop at ICML 2024	Vienna, Austria
10/2024	Invited talk. <i>Digital Twins of our Planet in the Era of Machine Learning.</i> AI+X summit	Zurich, Switzerland
09/2023	Contributed talk. <i>Spherical Fourier Neural Operators.</i> Large-scale Deep Learning for the Earth System workshop	Bonn, Germany
08/2023	Invited talk. <i>Data-driven weather prediction on the sphere.</i> Internal seminar at Environment and Climate Change Canada (ECCC)	virtual
05/2023	Contributed talk. <i>Spherical Fourier Neural Operators.</i> Tackling Climate Change with Machine Learning workshop at ICLR 2023	virtual
02/2018	Contributed talk. <i>Modeling tsunamis using the discontinuous Galerkin method for the spherical shallow water equations.</i> 2nd ASCETE workshop	Bayrischzell, Germany
06/2017	Contributed talk. <i>Large-Scale Tsunami Simulations using the Discontinuous Galerkin Method.</i> 27th Biennial Conference on Numerical Analysis	Glasgow, UK

Mentoring and Supervision

2023-now	Ankur Mahesh , collaborator (PhD student) co-supervised with William D. Collins; work published at GMD, and one paper in preparation	LBNL
2025-now	Franziska Gerken , Intern (PhD student) co-supervised with Laura Leal-Taixé; preprint in preparation	NVIDIA
2025-2025	Andrea Paris , Intern (Undergraduate student) work published at NeurIPS 2025	NVIDIA
2024-2025	Alberto Carpentieri , Intern (PhD student) paper under review	NVIDIA
2023-2024	Shoaib Saddiqui , Intern (PhD student) co-supervised with Kamyar Azizzadenesheli; paper under review	NVIDIA
2023-2024	Md Ashiqur Raman , Intern (PhD student) co-supervised with Kamyar Azizzadenesheli; work published at NeurIPS 2024	NVIDIA
2022-2023	Zilan Cheng , Master student co-supervised with Jan S. Hesthaven	EPFL
2018-2019	Mahya Hajihassanpour , Visiting PhD student co-supervised with Jan S. Hesthaven, work published in Ocean Modelling	EPFL
2016-2017	Lukas Prantl , Master student co-supervised with Nils Thuerey, work published at ICLR 2017	TUM

Teaching Experience

Spring 2020	Analysis 3 (for Physicists) , Principal Teaching Assistant	EPFL
Fall 2019	Linear Algebra (for Life Sciences) , Principal Teaching Assistant	EPFL
Spring 2019	Probability and Statistics , Principal Teaching Assistant	EPFL
Fall 2018	Probability and Statistics , Principal Teaching Assistant	EPFL
Spring 2018	Analysis 2 (for Electrical Engineers) , Principal Teaching Assistant	EPFL
Fall 2017	Analysis 1 (for Physicists) , Principal Teaching Assistant	EPFL
Spring 2017	Analysis 2 (for Physicists) , Principal Teaching Assistant	EPFL
Fall 2016	Seminar on Machine Learning in Graphics , Teaching Assistant	TUM

Awards and Honors

- 05/2023 **Work presented at the White House**, Presidential Council of Advisors in Science and Technology (PCAST), presented by A. Anandkumar 2023
- 07/2018 **Fellowship**, German National Scholarship Foundation (Studienstiftung des deutschen Volkes)

Professional Activities and Service

OPEN SOURCE CONTRIBUTIONS

- [torch-harmonics](#) – Python library for differentiable spherical harmonic transforms and models (~10k downloads/month)
- [makani](#) – Python library for GPU-accelerated weather and climate modeling
- [neuraloperator](#) – Python library for learning in infinite dimensions
- [HssMatrices.jl](#) – Julia library for hierarchically semi-separable (HSS) matrices
- [HierarchicalSolvers.jl](#) – Julia library for efficient quasilinear-complexity sparse direct solvers

PEER REVIEW

Neural Information Processing Systems (NeurIPS), International Conference on Machine Learning (ICML), International Conference on Learning Representations (ICLR), Advances in Computational Mathematics (AC), Quarterly Journal of the Royal Meteorological Society (QJMR), Nature Communications (Nat. Comms) and various workshops.

PROFESSIONAL MEMBERSHIPS

Society of Industrial and Applied Mathematics (SIAM), Association for Computing Machinery (ACM), European Laboratory for Learning and Intelligent Systems (ELLIS)

Other Experiences

PARENTAL LEAVE

3 months leave for the birth of my son

12/2024

COMPULSORY MILITARY SERVICE

9 months in the German Army

07/2008

Languages and Technical Skills

Theory	Machine Learning, Numerical & Functional Analysis, Numerical Linear Algebra, Information Theory
Code	Python, C++/C, CUDA, Julia, Fortran, Matlab, Mathematica
Libraries	PyTorch, numpy, JAX, TensorFlow, Triton, scipy, MPI, OpenMP, NCCL, matplotlib, pandas
Tools	git, slurm, make, Cursor, VSCode, Jupyter, Unix/Linux, cmake, bash, Paraview, $\LaTeX$
Languages	English (C2), German (C2), French (B2), Spanish (B2), Japanese (B1), Bulgarian (C2)